

Automated, on-demand Sentinel imagery retrieval through a service-based workflow

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How to make Sentinel imagery retrieval configurable, resource-efficient and accessible without requiring detailed Sentinel Hub expertise.

1 Introduction

We developed a configurable tool for the automated retrieval of Sentinel-1 and Sentinel-2 imagery for air- and water-pollution monitoring. Built on the Copernicus Data Space Ecosystem and Sentinel Hub, it supports user-defined areas, time periods, data products, spatial resolutions and quality thresholds. The tool also provides the imagery-retrieval layer of a broader workflow for identifying objects in aquatic environments

2 From example to operational tool

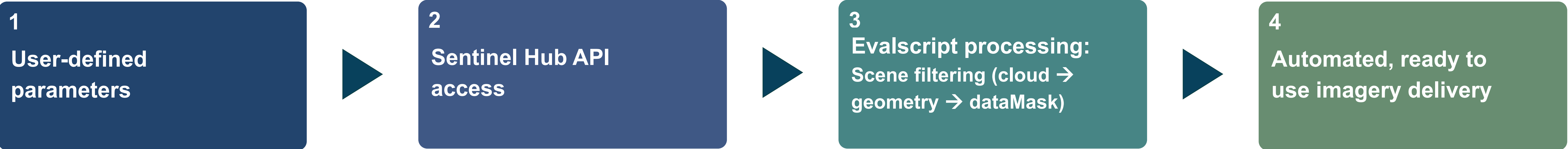
Built for Operational Use

- Extends Sentinel Hub's example evalscripts into a detailed, operational-grade query
- Eliminates hardcoding wherever possible
- Passes control to users unfamiliar with Sentinel Hub infrastructure
- Offers configurable parameters for essential query customisation

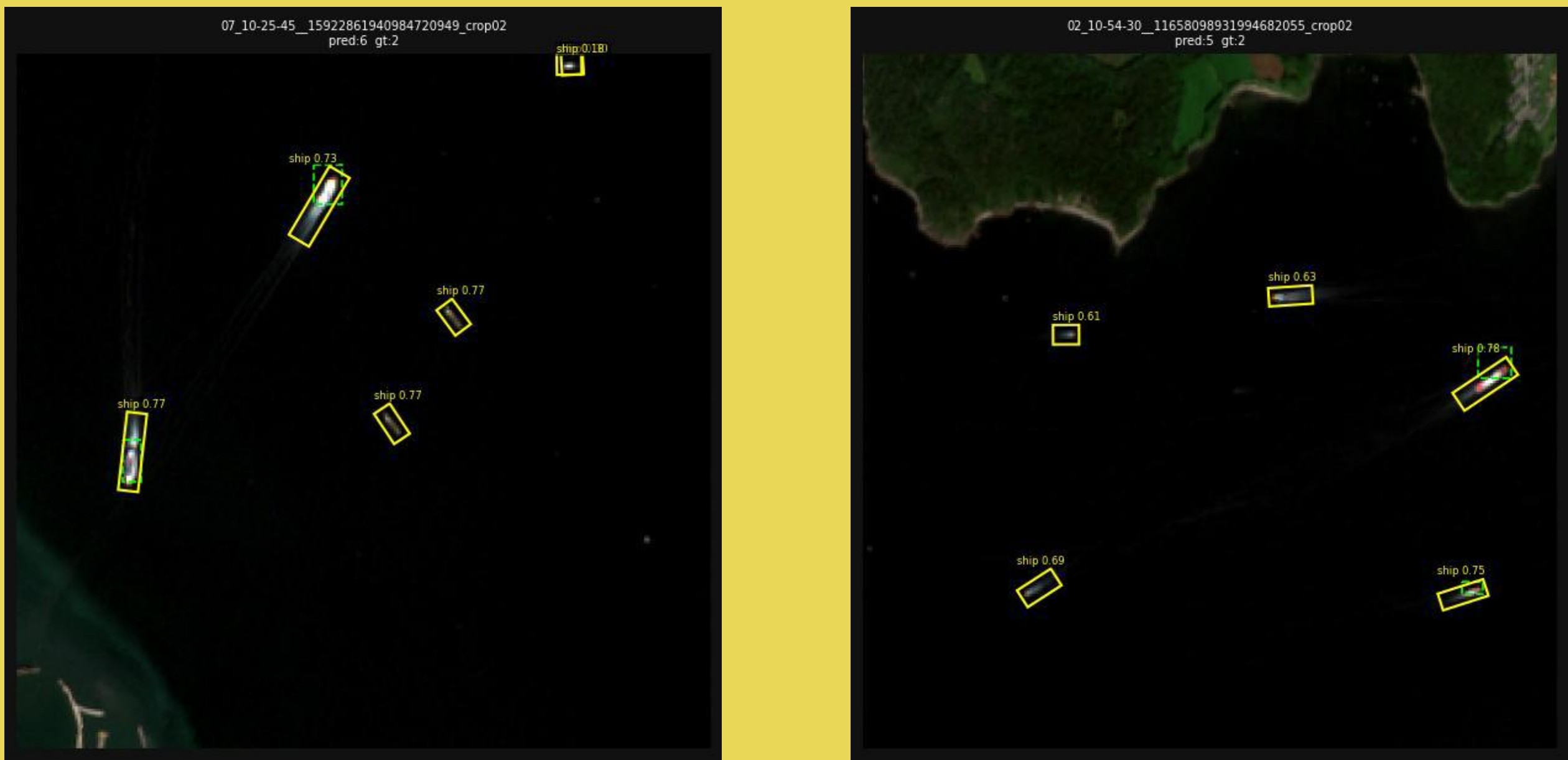
Efficient & Accessible by Design

- Three-stage filtering rejects unusable scenes *before* spending processing units on full-resolution retrieval
- Duplicate-download prevention skips re-fetching existing S1/S2 data for the same area and time
- Minimal, GDAL-free dependency set for reliable installation across platforms

3 Processing workflow



4 Applications & expected impact



The images above show examples of ship detection in the Baltic sea using YOLOv26

- Supports object identification research in aquatic environments
- Underpins emissions estimation studies
- Broadly applicable for any type of satellite image retrieval
- Lowers the barrier to entry for non-expert Earth Observation data users

5 Summary

The tool automates resource-efficient retrieval of Sentinel-1 and Sentinel-2 imagery through configurable queries, early scene filtering and duplicate-download checks. Its lightweight dependency stack simplifies installation and removes the need for detailed Sentinel Hub expertise. Ongoing work connects the retrieval module to aquatic-object detection and ship-emission estimation.